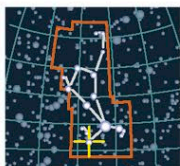


DOUBLE STAR

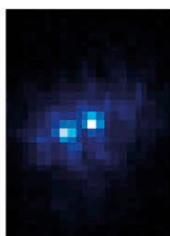
Zeta Boötis



BOOTES

DISTANCE FROM SUN
180 light-years
MAGNITUDE 3.8
SPECTRAL TYPE A3

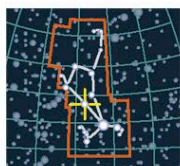
Zeta (ζ) Boötis would appear to be a standard double star—two A-type stars orbiting each other with a period of about 123 years. However, anomalies in calculations of its mass have suggested that there is something strange about the Zeta Boötis system. The answer lies in a highly elongated orbit, in which the stars range from 130–5,900 million miles (210–9,500 million km) apart. At their closest, they are almost as close as the Sun and Earth, and no telescope can visually split them. The Zeta Boötis system is about 40 times as luminous as the Sun, with about four times its mass, and has a temperature of about 15,700°F (8,700°C).



ENHANCED IMAGES
When the components of Zeta Boötis are at their farthest apart, image-processing software can be used to separate them and even split their spectra.

DOUBLE STAR

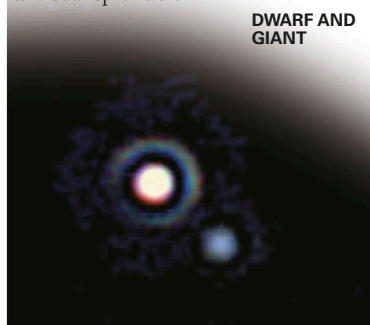
Izar



BOOTES

DISTANCE FROM SUN
210 light-years
MAGNITUDE 2.4
SPECTRAL TYPE A0

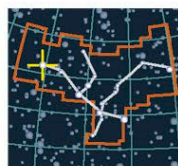
Izar, or Epsilon (ϵ) Boötis, is one of the best double stars in the sky. Its stars exhibit a striking color contrast—an orange giant close to a white dwarf—and it was given the name Pulcherrima, “most beautiful,” by its discoverer, German-born Friedrich Struve. The dwarf star is about twice the size of the Sun, while the orange giant is about 34 times the size. The dwarf and giant orbit each other with a period of more than 1,000 years. This double star is not particularly astronomically unusual but is well known to amateur astronomers for its visual splendor.



DWARF AND GIANT

QUADRUPLE STAR

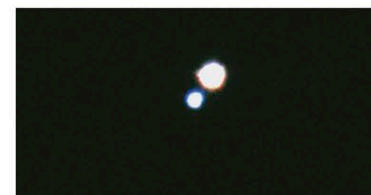
Almach



ANDROMEDA

DISTANCE FROM SUN
355 light-years
MAGNITUDE 2.3
SPECTRAL TYPE K3

Almach, or Gamma (γ) Andromedae, is well known to amateur astronomers as being a fine example of a double star with contrasting colors. The brighter star is yellow-orange, and the fainter star is blue; through a telescope, the two colors enhance

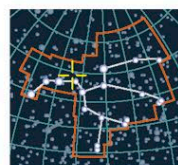


ALMACH

each other. The brighter star is a giant K-type star, while the fainter star is itself a double star, consisting of two hot, white main-sequence stars in a mutual orbit, with a period of about 60 years. It is difficult to split these two stars visually, but spectroscopic analysis reveals that one of them is also a double star in turn, making Almach a quadruple system.

DOUBLE STAR

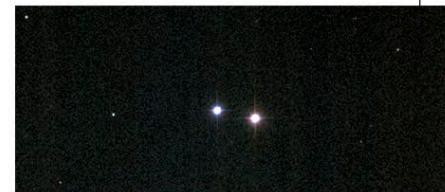
M40



URSA MAJOR

DISTANCE FROM SUN
1,900 and 550 light-years
MAGNITUDE 8.4
SPECTRAL TYPE G0

Some multiple stars are famous for their beauty, and others are famous for the dramatic astrophysics played out within the system. In the case of M40, neither applies. When compiling his well-known catalog of star clusters and nebulae, Charles Messier



OPTICAL PAIR

(see p.73) observed two stars close to each other in the night sky and erroneously included them. The two stars are nothing more than an optical double—that is, they happen to lie on the same line of sight. Modern distance measurements have shown that they are not truly associated with each other. M40 is therefore a double that achieves fame through error.

QUADRUPLE STAR

Alcyone



TAURUS

DISTANCE FROM SUN
368 light-years
MAGNITUDE 2.9
SPECTRAL TYPE B7

Alcyone, one of the sisters of the Pleiades (see p.291), is a bright giant star of spectral type B that shines about 1,500 times more brightly than the Sun. Orbiting around Alcyone are three stars forming a compact system: 24 Tau (magnitude 6.3) and V647 Tau (magnitude 8.3) are both A-type stars, while HD 23608 (magnitude 8.7) is an F-type star. V647 Tau is a variable of the Delta Scuti type. The system of three stars orbits Alcyone at a distance

of a few billion miles. Alcyone is unusual in that it rotates at high speed. This has caused it to throw gas off at its equator, which forms a light-emitting disk. It is classified as a Be star (see p.285), similar to Gamma (γ) Cassiopeiae.

SEASONAL SIGNAL

Alcyone is the brightest star in the Pleiades Cluster (see p.291). Its appearance over the eastern horizon signals the start of fall in the Northern Hemisphere.